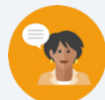


7.5 Using Trigonometric Ratios

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Targets	- I can use trigonometric ratios to find the side of a right triangle. - I can use trigonometric ratios to find an angle of a right triangle.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How can I use trigonometric ratios to find the side of a right triangle? - How can I use trigonometric ratios to find an angle of a right triangle?			
 Lesson Narrative	In this lesson, students build on the previous lessons by using trigonometric ratios to determine the value of missing sides. They will then be introduced to using inverse trigonometric functions to find the missing angle measure in a right triangle and how to use trigonometric ratios to find missing side measures of a right triangle.			
 Component Summary	7.5.1 Warm-Up (~5 min.)	Activate prior knowledge (<i>SE p. 394</i>)	7.5.4 Guided Instruction (10 min.)	Use trigonometric ratios to find measures (<i>SE p. 397</i>)
	7.5.2 Collaboration (10 min.)	Use trigonometric ratios to write equations (<i>SE p. 394</i>)	7.5.5 Guided Practice (5 min.)	Use trigonometric ratios to find measures (<i>SE p. 398</i>)
	7.5.3 Guided Instruction (10-15 min.)	Use inverse trigonometric functions to find angle measures (<i>SE p. 396</i>)	7.5.6 Wrap-Up (~5 min.)	Find the measure of a missing side and a missing angle (<i>SE p. 399</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Inverse Trigonometric Functions - Arccosine - Arcsine - Arctangent <u>Additional Terms:</u> N/A		Scientific Calculator	N/A

 Teacher Tips	Common Misconceptions - Students may use trigonometric functions to find missing angle measures instead of inverse trigonometric functions. - Students may incorrectly use their calculator with trigonometric function keys when completing calculations.	Things to Consider Students have learned about trigonometric ratios in the prior lesson. In this lesson, students will be using them to solve problems. Some students may need remediation on the basics of trigonometric functions.
	Literacy and Language Routines Compare and Connect This lesson consists of two unique activities that are teacher led, 7.5.3 and 7.5.4 Guided Instructions. During this process, the opportunity for teachers to share their thinking out loud will be available. Students will be prompted to reflect and respond. This routine supports meta-cognitive and meta-linguistic awareness and also supports mathematical conversation.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Throughout this lesson, students will use patterns and structures to help understand and connect mathematical concepts. Students will decompose complex problems into manageable parts and relate previously learned concepts in earlier lessons to new concepts in this lesson.
	 Differentiate Instruction	
Consider the creation of a foldable that provides a difference between the use of trigonometric functions to solve a problem (one for each function) and using inverse functions to solve problems (one for each function), so students can have a reference guide for the two scenarios. Additional differentiation opportunities are denoted throughout the lesson guide.		
Lesson Notes:		

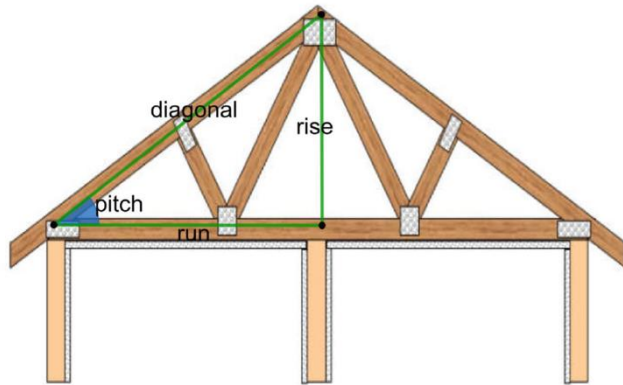
7.5.1 Warm-Up: Notice and Wonder

Component Overview: Students activate prior knowledge of right triangles through a Notice and Wonder.



7.5.1 Warm-Up: Notice and Wonder

In carpentry, right triangles are commonly used in constructing roofs. A roof's pitch is the steepness of the roof. It can be determined by the ratio between the two legs of the right triangle.



Prompt students to make the connection between this question and the Pythagorean Theorem, which we have been working on in the last few lessons.

1. What do you notice or wonder about the use of right triangles in construction?

Answers will vary. I notice that the triangles used in the picture are equilateral and are right triangles. I wonder what other types of triangles are commonly used in carpentry.

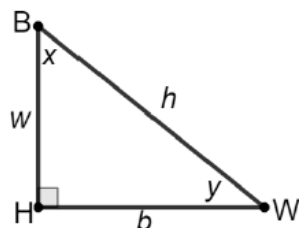
7.5.2 Collaboration: Using Trigonometric Ratios

Component Overview: Students will work together to write equations using trigonometric ratios.



7.5.2 Collaboration: Using Trigonometric Ratios

1. $\triangle BWH$ is shown, where
 $h = 12.8$, $w = 8$,
 $x = 51.3^\circ$, and $y = 38.7^\circ$.



- a. Select all of the equations that can be used to find the value of b .

☐ $\cos(51.3^\circ) = \frac{8}{b}$ ☐ $\sin(51.3^\circ) = \frac{b}{12.8}$ ☐ $\tan(51.3^\circ) = \frac{b}{12.8}$

☐ $\cos(38.7^\circ) = \frac{b}{12.8}$ ☐ $\sin(38.7^\circ) = \frac{b}{12.8}$ ☐ $\tan(38.7^\circ) = \frac{8}{b}$

- b. Determine the value of b . Round to the nearest thousandth.

$$\sin(51.3^\circ) = \frac{b}{12.8}$$

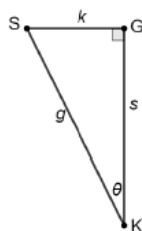
$$b = 9.990$$

2. $\triangle SGK$ is shown.

- a. Use a trigonometric ratio for θ to write an equation that can be used to find the value of k .

$$\sin(\theta) = \frac{k}{g}$$

$$\tan(\theta) = \frac{k}{s}$$



Student responses to question 1a can provide formative data to determine which students may need additional support to remediate the information learned in the prior lessons in this unit.

7.5.2 Collaboration: Using Trigonometric Ratios (continued)

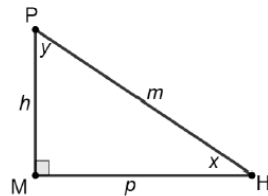
- b. Which trigonometric ratio for θ cannot be used to write an equation that can be used to find the value of k ?

$$\cos(\theta) \text{ since } \cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{s}{g}.$$



How did you determine which trigonometric ratio could be used to determine the value of k ?

3. $\triangle PMH$ is shown. Work with your partner to determine which trigonometric ratio can be used to determine the unknown side or angle. Then, use the ratio to write an equation that can be used to determine the unknown. When writing the equation, use the variables h , p , m , x , and y .



Given	Unknown	Trigonometric Ratio	Equation
$m\angle PHM$ and length of $\overline{PM}$	$\overline{PH}$	☐ sine ☐ cosine ☐ tangent	$\sin(x) = \frac{h}{m}$
$m\angle PHM$ and length of $\overline{PM}$	$\overline{MH}$	☐ sine ☐ cosine ☐ tangent	$\tan(x) = \frac{h}{p}$
$m\angle HPM$ and length of $\overline{PM}$	$\overline{PH}$	☐ sine ☐ cosine ☐ tangent	$\cos(y) = \frac{h}{m}$
Lengths of $\overline{MH}$ and $\overline{PH}$	$m\angle HPM$	☐ sine ☐ cosine ☐ tangent	$\sin(y) = \frac{p}{m}$
Lengths of $\overline{MH}$ and $\overline{PM}$	$m\angle PHM$	☐ sine ☐ cosine ☐ tangent	$\tan(x) = \frac{h}{p}$

7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure

Component Overview: Students will be introduced to using the inverse of trigonometric functions to find angle measures through three multi-part questions.



7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure

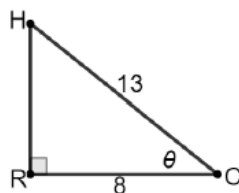


Inverse Trigonometric Functions

$\cos^{-1}\left(\frac{a}{c}\right)$ is arccosine. $\sin^{-1}\left(\frac{b}{c}\right)$ is arcsine. $\tan^{-1}\left(\frac{b}{a}\right)$ is arctangent. These functions can be used to determine the measure of an angle where a is the side adjacent the angle, b is the side opposite the angle, and c is the hypotenuse. The calculator should be in degree mode.

1. In $\triangle HRC$, $m\angle HRC = 90^\circ$, $RC = 8$, and $HC = 13$.

The equation $\cos(\theta) = \frac{8}{13}$ can be used to determine θ . To find θ using a calculator use the $\cos^{-1}\left(\frac{8}{13}\right)$. Round any angle measurements below to the nearest whole degree.



- a. What is $\cos^{-1}\left(\frac{8}{13}\right)$?

$$\cos^{-1}\left(\frac{8}{13}\right) \approx 52^\circ$$

- b. Write an equation that can be used to find $m\angle CHR$.

$$m\angle CHR = \sin^{-1}\left(\frac{8}{13}\right)$$

- c. Find $m\angle CHR$ using an inverse trigonometric function on a calculator. Be sure the calculator is in degree mode.

$$m\angle CHR = \sin^{-1}\left(\frac{8}{13}\right) \approx 38^\circ$$



Consider the creation of a foldable that provides a difference between the use of trigonometric functions to solve a problem (one for each function) and using inverse functions to solve a problem (one for each function), so students can have a reference guide for the two scenarios.



Students may need additional support with inputting the inverse trig function into their calculator. Consider modeling with a digital calculator, such as Desmos scientific calculator, or using a document camera, to show students how to input the values into their calculators. If needed, consider giving them a few to try to ensure students all get the same answer.



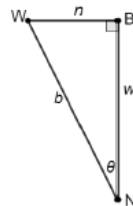
Why can this equation be used to determine the value of the unknown angle? Could another one be used? Why or why not?

7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure (continued)

2. $\triangle WLT$ is a right triangle with $m\angle WTL = 90^\circ$, $TL = 11$, and $WL = 23$. What is the $m\angle TWL$? Round to the nearest whole degree.

$$m\angle TWL = \sin^{-1}\left(\frac{11}{23}\right) \approx 29^\circ$$

3. $\triangle WBN$ is shown, where $\angle B$ is a right angle.



- a. Determine the measure of $\angle BNW$ if $n = 36$ and $w = 77$.

$$\angle BNW = \tan^{-1}\left(\frac{36}{77}\right) \approx 25^\circ$$

- b. Gabriella used $\sin^{-1}\left(\frac{77}{85}\right)$ to find the measure of $\angle BNW$. Determine if $\sin^{-1}\left(\frac{77}{85}\right)$ results in the same degree measure as your answer.

No, it does not:

$$\sin^{-1}\left(\frac{77}{85}\right) \approx 65^\circ.$$

- c. Discuss with your partner whether or not Gabriella's method is valid. Write a justification of your decision.

Answers may vary.

Gabriella's method is not valid. This is because she set up her arcsine formula incorrectly. Sine (or arcsine) is always $\frac{\text{opposite}}{\text{hypotenuse}}$, however, she used $\frac{\text{adjacent}}{\text{hypotenuse}}$, which is the cosine. It is due to this that resulted in Gabriella getting the incorrect value for the angle measurement.



At the conclusion of this activity, engage students in a discussion around their responses and thinking for questions 3b and 3c. This provides an opportunity to uncover potential misconceptions and provides students the opportunity to articulate their thinking and learn from their peers.

7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures

Component Overview: Students will use trigonometric ratios to find side measures through three, multi-part questions.



7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures

1. In $\triangle KDR$, $m\angle DKR = 31.2^\circ$, $m\angle KDR = 90^\circ$, and $KD = 17$. What is the length of $\overline{KR}$?

The trigonometric ratio to find the approximate length of KR will be cosine, since $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$.

$$\cos(31.2^\circ) = \frac{17}{KR}$$

$$KR \cdot \cos(31.2^\circ) = 17$$

$$KR = \frac{17}{\cos(31.2^\circ)}$$

$$KR \approx 19.875$$

2. $\triangle MAU$ is a right triangle with $MA = 20.7$ and $MU = 9.1$. If $m\angle MUA = 90^\circ$, what is the $m\angle AMU$?

The trigonometric ratio to find $m\angle AMU$ will be cosine and arcsine, since $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. $\cos(\theta) = \frac{9.1}{20.7}$, which means that $\cos^{-1}\left(\frac{9.1}{20.7}\right) = \theta = m\angle AMU$.

$$\cos^{-1}\left(\frac{9.1}{20.7}\right) \approx 63.921^\circ$$

3. $\triangle NKD$ is shown, where $\angle K$ is a right angle, $KD = 36$, and $m\angle D = 65^\circ$

- a. Find the length of $\overline{ND}$. Show your work.

$$\cos(65^\circ) = \frac{36}{ND}$$

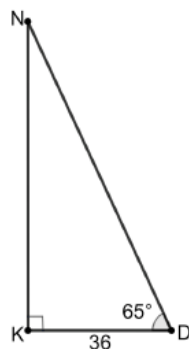
$$ND \cdot \cos(65^\circ) = \frac{36}{ND} \cdot ND$$

$$ND \cdot \cos(65^\circ) = 36$$

$$ND \cdot \frac{\cos(65^\circ)}{\cos(65^\circ)} = \frac{36}{\cos(65^\circ)}$$

$$ND = \frac{36}{\cos(65^\circ)}$$

$$ND \approx 85.183$$



This lesson consists of two unique guided instruction activities. During this process, the opportunity for teachers to share their thinking out loud will be available. Students will be prompted to reflect and respond. This routine supports meta-cognitive and meta-linguistic awareness and also supports mathematical conversation.



How did you determine which trigonometric function would help determine the angle in question 2?



How did you determine which trigonometric function would help determine the length of the side in question 3a?

7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures (continued)

- b. Amelia and Henry found the length of $\overline{NK}$ using two different methods. The first step of their work is shown. Complete their work.

Amelia	Henry
$36^2 + NK^2 = (85.183)^2$	$\tan(65^\circ) = \frac{NK}{36}$
$1296 + NK^2 = 7256.1435$	
$1296 - 1296 + NK^2 = 7256.1435 - 1296$	$36 \cdot \tan(65^\circ) = \frac{NK}{36} \cdot 36$
$NK^2 = 5960.1435$	$36 \cdot \tan(65^\circ) = NK$
$\sqrt{NK^2} = \sqrt{5960.1435}$	
$NK \approx 77.20197$	$NK \approx 77.20225$

- c. Describe some possible reasons that Amelia's and Henry's answers are not exactly the same value.
Answers will vary.
Amelia used a rounded value for the hypotenuse.
- d. Whose method gives a precise answer?
Henry's answer is more precise.



Before moving to the next component, address any misunderstandings or clarifications that arise during 7.5.4 Guided Instruction.

7.5.5 Guided Practice: Using Trigonometric Ratios

Component Overview: Students will practice using trigonometric ratios to find missing angles and sides using two questions.



7.5.5 Guided Practice: Using Trigonometric Ratios

1. $\triangle ZME$ has $ME = 14.9$ and $m\angle ZEM = 41.5^\circ$.

- a. If $\angle ZME$ is a right angle, what is the length of $\overline{ZM}$?

$$\begin{aligned}\tan(41.5^\circ) &= \frac{ZM}{14.9} \\ 14.9 \cdot \tan(41.5^\circ) &= ZM\end{aligned}$$

$$ZM \approx 13.182$$

- b. Sharif used the information to determine the length of $\overline{ZM}$.
Her work is shown below.

$$\begin{aligned}\tan 41.5^\circ &= \frac{14.9}{ZM} \\ ZM \cdot \tan 41.5^\circ &= 14.9 \\ ZM &= \frac{14.9}{\tan 41.5^\circ} \\ ZM &\approx 16.841\end{aligned}$$

Find Sharif's error and correct it.

Sharif wrote the ratio of tangent as $\frac{\text{adjacent}}{\text{opposite}}$ instead of $\frac{\text{opposite}}{\text{adjacent}}$.

$$\begin{aligned}\tan 41.5^\circ &= \frac{ZM}{14.9} \\ 14.9 \cdot \tan 41.5^\circ &= ZM \\ ZM &\approx 13.182\end{aligned}$$

2. $\triangle KNP$ is a right triangle with $KP = 46$ and $KN = 27.3$. If $\angle PNK$ is a right angle, what is the $m\angle KPN$?
Since the opposite side length and the hypotenuse are given, arcsine can be used to find the angle measure.

$$m\angle KPN = \sin^{-1}\left(\frac{27.3}{46}\right) \approx 36.404^\circ$$



This activity provides the only practice opportunity with this material for this concept. Consider extending the practice opportunity or using one question as an independent practice opportunity prior to the Ticket Out the Door in the next activity.



- Why is the statement "is a right angle" critical to solving this problem?
- Could you still use this method if it was not? Why or why not?

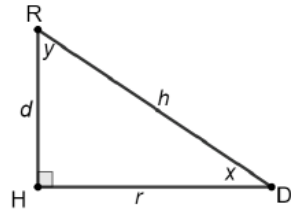
7.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a Ticket Out the Door that has students find a missing angle and a missing side using trigonometric functions. Students should be allowed to access a calculator to complete the wrap-up.



7.5.6 Wrap-Up: Ticket Out the Door

1. $\triangle RDH$ is shown, where $\angle RHD$ is a right angle.



- a. If $d = 2.8$ and $h = 4.6$, what is the measure of y , in degrees?

$$y = \cos^{-1}\left(\frac{2.8}{4.6}\right) \approx 52.505^\circ$$

- b. If $x = 29^\circ$ and $h = 14.5$ what is the value of d ?

$$\begin{aligned}\sin(29^\circ) &= \frac{d}{14.5} \\ 14.5 \cdot \sin(29^\circ) &= d \\ d &\approx 7.03\end{aligned}$$



Student responses to this activity can provide data to which components of the lesson, if any, may need additional support and help to distinguish what students need additional assistance with understanding the content.